



Automatic Diagnosis and Classification of Glaucoma Using Hybrid Features and k -Nearest Neighbor

Kishore Balasubramanian^{1,*}, N. P. Ananthamoorthy², and K. Gayathridevi³

¹EEE, Dr. Mahalingam College of Engineering and Technology, Pollachi 642002, India

²EEE, Hindusthan College of Engineering and Technology, Coimbatore 641032, India

³ECE, Dr. NGP Institute of Technology, Coimbatore 641048, India

Glaucoma detection is the most challenging aspect in the medical image processing and analysis field. In this paper, glaucoma detection is assessed by employing High-Resolution Fundus (HRF) images and RIMONE databases. The proposed approach contains four major steps: segmentation, feature extraction, dimensionality reduction and classification. At first, segmentation was carried-out using correlation based template matching, it was a flexible high level machine learning technique to localize the object in complex template. Secondly, hybrid feature extraction (homogeneity and correlation) performed on the segmented optic disc image after performing Haar Discrete Wavelet Transform (HDWT) in order to achieve feature subsets. The respective feature values were given as the input for Principal Component Analysis (PCA) for the rejection of irrelevant and redundant features. After finding the optimal feature information, a multi-objective classifier: K -Nearest Neighbor (KNN) employed for classifying the normality and abnormality of glaucoma disease. In experimental analysis, the proposed approach distinguishes the normality and abnormality of glaucoma disease by means of specificity, sensitivity, accuracy, Positive Predictive Value (PPV), Negative Predictive Value (NPV) and Matthews Correlation Coefficient (MCC). The experimental outcome showed that the proposed methodology improved accuracy in glaucoma detection up to 3–30% compared to the existing methods: Neural Network (NN), Naïve Bayes (NB), Random Forest (RF) and Support Vector Machine (SVM).

Keywords: Correlation, Haar Discrete Wavelet Transform, k -Nearest Neighbor, Principal Component Analysis, Template Matching.

1. INTRODUCTION

In recent decades, glaucoma detection is an emerging research topic in the fields of health monitoring and medical image analysis.^{1,2} Generally, glaucoma detection is a challenging research field, because the real world glaucoma images formed by the interaction of several factors in different conditions such as, illumination changes, background interference, lighting variation, etc.³ The human ability of identifying and recognizing the optic discs in glaucoma image is naturally significant, but it may not be precise all the time. Manual recognition consumes more time for segmenting the optic discs and also it is applicable only for a limited number of glaucoma images.^{4,5} So, the automatic glaucoma detection methodology has created an attention within the research community due to its several functional applications.^{6,7} In automatic glaucoma detection system, segmentation and classification are the two key steps.⁸ In order to

develop an effective glaucoma detection system, some of the concerns need to be resolved, especially on glaucoma classification.⁹ In classification section, the individual glaucoma images converted into vectors that are highly dimensional in nature. It is necessary to transform the high dimensional image space into low dimensional image space.^{10,11} Currently, a lot of dimensionality reduction methodologies are there in glaucoma detection to decrease “curse of dimensionality.”¹² In that, some common methodologies are Linear Discriminative Analysis (LDA), Independent Component Analysis (ICA), and so on.¹³

Performances of these methodologies highly affected by the number of samples in the training set.¹⁴ Particularly, when the number of samples is low, the learned sub-space projection might be deficient.¹⁵ In order to enhance the optic disc segmentation and classification, a new methodology was developed. In this experimental analysis, glaucoma images acquired from two datasets: HRF and RIMONE. After the acquisition of medical images, normalization was performed, which transformed the range of pixel intensity values for enhancing the quality

*Author to whom correspondence should be addressed.

of glaucoma image by eliminating the unwanted noise. After pre-processing the glaucoma images, correlation based template matching segmentation and blood vessels in-painting methods employed for segmenting the optic discs and to remove the unwanted blood vessels in optic discs. The correlation based template matching algorithm is a high level machine learning approach for localizing the optic discs in complex template. Then, an effective transformation methodology named as HDWT was employed, which helped to reveal the local characteristics and also helped in the reduction of feature degradation. Additionally, hybrid feature extraction (homogeneity and correlation) was employed for extracting the feature values. Also, these feature extraction methods helped to reflect the depth and smoothness of glaucoma image texture structure. The obtained feature values were given as the input for dimensionality reduction method: PCA for rejecting the irrelevant and redundant features. After obtaining the optimal feature information from PCA, the classification of normality and abnormality of the glaucoma was recognized by employing KNN classifier.

This paper is composed as follows. Section 2 surveys several recent papers on glaucoma detection related strategies. In Section 3, hybrid texture feature extraction methods presented with PCA and multi objective classification method: KNN. The Section 4 shows the comparative experimental result for existing and proposed strategy. The conclusion is made in Section 5.

2. LITERATURE REVIEW

Several researches suggested by researchers in glaucoma detection. In this scenario, a brief evaluation of some important contributions to the existing literatures is presented.

Mittapalli and Kande,¹⁶ proposed a new implicit region based active contour methodology for optic disk segmentation. The proposed methodology integrated the information about the image at the point of interest from several image channels to have effectiveness against the variations found in and around the optic disk region. In this literature, the proposed approach: Spatially Weighted Fuzzy C Means (SWFCM) clustering technique was based on the gray level and structural properties of cup. The obtained information about the optic disk contours used to calculate the cup different parameters for glaucoma assessment. Experiments were carried out on the RIMONE and DIARETDB0 databases for demonstrating the robustness of the proposed method. The disadvantage of SWFCM methodology was more complex to find the projection of free space. Free space can able to provide better discriminant ability in feature extraction.

Salam et al.¹⁷ evaluated a new algorithm for detecting glaucoma from digital fundus image using a hybrid feature set. The proposed algorithm was the combination of both structural (cup to disc ratio) and non-structural (intensity and texture) features for enhancing the efficiency of automated diagnosis of glaucoma. In addition, the proposed approach developed a suspect class in non-structural and structural features for automatic diagnosis. Experimental outcome confirmed that the proposed approach outperformed the existing method in terms of sensitivity and specificity. The hybrid feature set was highly affected by the number of samples in the training set. When the number of samples becomes low, the detection rate minimized.

Roychowdhury et al.¹⁸ presented an effective classification-based optic disc segmentation methodology for detecting optic disk boundary and location of the vessel origin pixel. At first,

the green plane of each fundus images was morphologically reconstructed and resized using circular structuring element. Then, the bright regions were extracted from the morphologically reconstructed image, which lies close to the major blood vessels. Finally, the extracted regions were classified by using Gaussian mixture model classifier. The experiment was carried on publicly available databases: DIARETDB1, DIARETDB0, DRIVE, MESSIDOR, CHASE DB1, and STARE to validate its accuracy, robustness and speed. In some cases, the training data were dependent evaluation or manual adjustment, which is needed to be automated.

Maheshwari et al.¹⁹ developed a new methodology for automatic glaucoma detection using digital fundus images. In this literature, Variational Mode Decomposition (VMD) methodology was utilized as an image decomposition algorithm. Then, feature extraction approaches like Kapoor, yager, and Renyi entropy were utilized for extracting the feature values from VMD components. Finally, least-squares support vector machine classification methodology was employed to classify the extracted regions. Experimental outcome showed that the proposed glaucoma detection methodology was very effective compared to the existing methodologies. While performing with semi supervised methodologies, the semantic gap was maximized between the feature values, which leads to poor detection rate.

Singh et al.²⁰ presented a new methodology: wavelet feature extraction and optimized genetic feature selection with numerous learning methodologies. This research work was performed using feature extraction on segmented blood vessels and optic disc for improving the detection accuracy. Validated outcome confirmed that the proposed methodology was very effective compared to the existing schemes. In this research, the fixed thresholding was not adequate to handle large intensity variations in the cup region that arise due to physiological difference across patients.

Wang et al.²¹ proposed a novel intelligent emotion recognition system using stationary wavelet entropy and a single hidden layer feedforward neural network. Initially, the stationary wavelet entropy extracts the features from facial images and classifies the extracted feature vectors using single hidden layer feedforward neural network. To prevent the training of the classifier, fall and to obtain local optimum points, a new algorithm was introduced named as jaya algorithm. The simulation result was analyzed using 700 image dataset and the developed approach achieved the overall accuracy of $96.80 \pm 0.14\%$. In stationary wavelet entropy algorithm, there was a need to identify Eigen values and Eigen vector that leads to time consumption.

Zhang et al.²² proposed stationary wavelet transform (SWT) approach for extracting the features from MR brain images. Then, principal component analysis (PCA) was employed to reduce the SWT coefficients. Finally, two new classifiers were proposed; Generalized Eigenvalue Proximal Support Vector Machine (GEPSVM), and GEPSVM with RBF kernel for classifying the optimal feature vectors. The experimental analysis was tested on three benchmark datasets: dataset-66, dataset-160, and dataset-255, respectively. The proposed method achieved accuracy of 100%, 100%, and 99.41% on dataset-66, dataset-160, and dataset-255. Though, PCA was highly affected by the number of samples per person in the training set, when the number of samples were low that affects the recognition rate.

Lu et al.²³ proposed a new pathological brain detection system using 2D discrete wavelet transform (DWT) and Radial Basis

Function Neural Network (RBFNN). The DWT extracts the features from brain images and the extracted feature vectors were classified using RBFNN. A 10×10 fold cross validation was conducted to prevent overfitting. The method achieved a sensitivity of 95.89%, a specificity of 92.78%, and an overall accuracy of 95.44% over 125 MR brain images. The performance suggested that the proposed classifier was robust and effective in comparison with recently state-of-the-art approaches. In large dataset, RBFNN failed to achieve better detection by means of accuracy.

To overcome the above mentioned drawbacks, an unsupervised algorithm (combination of hybrid texture feature extraction with multi objective classification) is implemented for enhancing the performance of glaucoma prediction.

3. PROPOSED METHODOLOGY

The proposed glaucoma detection system consists of six steps such as, image acquisition, image pre-processing, optic disc segmentation and blood vessel removal, feature extraction, dimensionality reduction and classification. The proposed glaucoma detection system block diagram is represented in the Figure 1. The brief description about the proposed technique is described below.

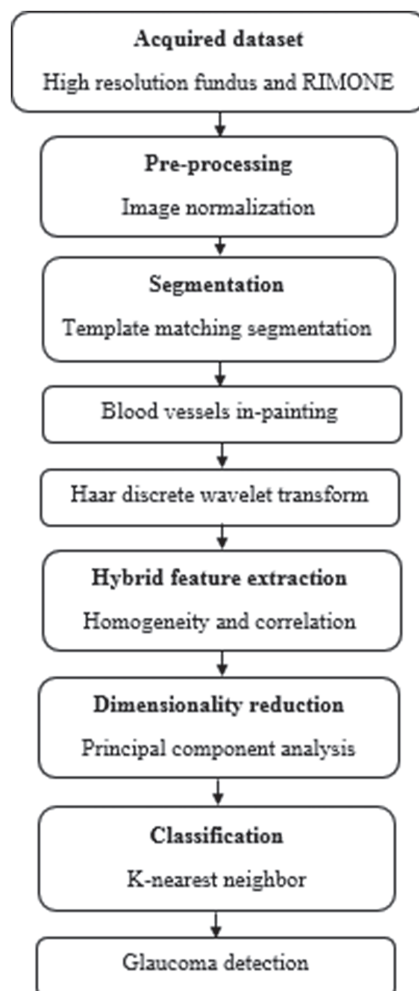


Fig. 1. Block diagram of proposed methodology.

3.1. Database Acquisition

In the initial stage of glaucoma detection system, the glaucoma images are taken from the standard benchmark datasets: HRF and RIMONE database. HRF database contains healthy patients 15 images, diabetic retinopathy patients 15 images and glaucomatous patients 15 images. For each image, binary gold standard vessel segmentations are available. The binary gold standard data are created by a group of clinicians working in the field of retinal image analysis and experts from ophthalmology clinics. In addition, RIMONE database contains 169 high-resolution images, each image have 5 manual reference segmentation images. These images decreases the variability among clinicians or expert segmentation and also helps to achieve high accurate segmentation algorithms. The sample glaucoma images are denoted in the Figure 2.

3.2. Pre-Processing Using Normalization

After image acquisition, a pre-processing method: normalization is performed on the glaucoma image for enhancing or de-noising the fundus image. Most of the fundus retinal images are collected or captured from fundus equipment's, whereas the collected retinal images are majorly consisting of two noises: impulse noise and machinery noise (electrical and mechanical noises). The normalization methodology is very effective to remove impulse and machinery noise, also helps to enhance the image quality significantly. Normalization is a procedure that modifies the range of pixel intensity values for enhancing the quality of glaucoma image by eliminating noise from the images. Then, determine the alternations and deformation occurred in the glaucoma image by inaccurate image capture. The sample normalized glaucoma image is represented in the Figure 3.

Image normalization consists of pixel intensity variations. The entire glaucoma image is transformed into pre-defined values, because it is a pixel wise process. Suppose, the intensity range is m to n , and the desired range is p to q . In this condition, initial step in normalization methodology is subtracting m from each pixel intensity. The general formula of image normalization methodology is determined in the Eq. (1).

$$IN = (I - \text{Min}) + \frac{\text{newMax} - \text{newMin}}{\text{Max} - \text{Min}} + \text{newMin} \quad (1)$$

Where, I is represented as original glaucoma image, IN is stated as new image, ($\text{Min} = 0$, $\text{Max} = 255$) is specified as intensity range of original image and (newMin , newMax) is stated as intensity range of new image.

3.3. Optic Disc Segmentation and Blood Vessels Removal

The pre-processed glaucoma images used for optic disc segmentation using correlation based template matching segmentation.

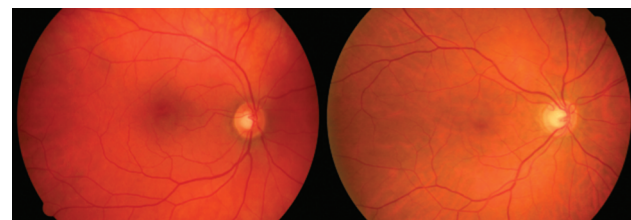


Fig. 2. Sample glaucoma images.

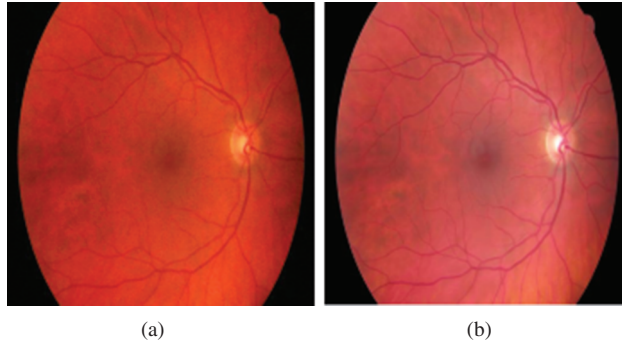


Fig. 3. (a) Noisy image. (b) Normalized image.

Template matching segmentation is the procedure of localizing the positions in which a specified template located inside a glaucoma image. The correlation based template matching procedure involves three steps;

- Moving the template with-in the search area.
- Compute, similarity between the glaucoma positioned image area and template for each template location.
- Determine the positions, where a good similarity measure achieved.

Finally, the correlation between the image window and template used as a measure of similarity in glaucoma image registration and template matching. The graphical representation of template matching represented in the Figure 4.

To further improve the accuracy of optic disc detection, the blood vessels around optic discs should be removed. In this scenario, the blood vessels removed by employing an algorithm, which is based on the pixel intensity function of the distance from center of the optic discs. The gray level pixel value reduces the distance from the optic disc center. The pixels that have equal distance from the optic disc center have same intensity values. The remaining deviation patterns removed, which represented as blood vessels. The algorithm of blood vessel in-painting denoted below.

Step 1: Consider optic disc image I and center of optic disc (x_0, y_0) .

Step 2: Let, $p = \{p_1, p_2, p_3 \dots p_k\}$ is the set of pixels, which is equal to the distance r from the optic disc center.

Step 3: Calculate,

$$T = \min(p) + \frac{\{\max(p) - \min(p)\}}{2} \quad (2)$$

Step 4: If $p_q < T$, then p_q is determined as blood vessel pixel.

Step 5: Remove the blood vessel pixels p_q . Then, the pixel value is replaced by the median of the pixels of p using $p_{q(\text{new})} = \text{median}(p_1, p_2, p_3, p_k)$ After optic disc segmentation, wavelet

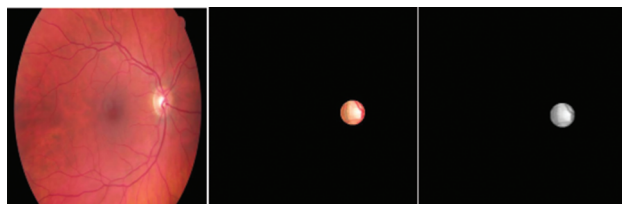


Fig. 4. Template matching based segmentation.

transformation carried-out on the optic disc segmented image after blood vessels in-painted, which is graphically stated in the Figure 5.

3.4. Haar-Discrete Wavelet Transform

The HDWT is an effective tool for image processing applications and it adopted in several emerging standards. The Haar wavelet is a rescaled sequence “square-shaped” functions, which forms a wavelet family. The growing success of HDWT in the field of glaucoma detection is due to multiresolution processing capabilities and also huge range of filters used to improve image quality. The HDWT has properties like good compression of energy, perfect reconstruction with short support filters, no redundancy and very low computation. Also, the HDWT used as a fuzzy denoising procedure that provides shift able sub-bands and better directional selectivity with low redundancy.

In the multiresolution process, the original texture of glaucoma images evaluated by zooming in and out procedure. Initially, HDWT decomposes the glaucoma images into a number of sub-images in different resolution levels preserving the high and low frequency information. This property helps the wavelet to extract the information of texture from glaucoma images. The square integrable function $f(x)$ and its wavelet transform determined as the inner product f and a real valued function $\Psi(x)$ is represented in the Eq. (3).

$$w[f(s, \tau)] = (f, \Psi_{s,t}^k) = \int_{-\infty}^{\infty} f(x) \Psi_{s,t}^k(x) dx \quad (3)$$

Where, $\Psi_{s,t}^k(x) = (1/\sqrt{s})\Psi_{s,t}^k(x - \tau)/s$ is represented as wavelet family, $s \in \mathbb{Z}$, τ and $k \in \{h, v, d\}$ are denoted as scale (resolution level), translation and orientation parameters. The orientation parameters h , v , and d are represented as horizontal, vertical and diagonal directions respectively. Now, the dyadic wavelet decomposition is achieved, when $s = 2^j$ and $\tau = 2^j n$, $j, n \in \mathbb{Z}$. Using the wavelet function $\Psi(x)$ and the scaling function $\varphi(x)$, the wavelet and scaling families are constructed using the Eqs. (4) and (5).

$$\Psi_{j,n}^k(x) = \frac{1}{\sqrt{2^j}} \Psi^k\left(\frac{x - 2^j \cdot n}{2^j}\right) \quad (4)$$

$$\varphi_{j,n}^k(x) = \frac{1}{\sqrt{2^j}} \varphi^k\left(\frac{x - 2^j \cdot n}{2^j}\right) \quad (5)$$

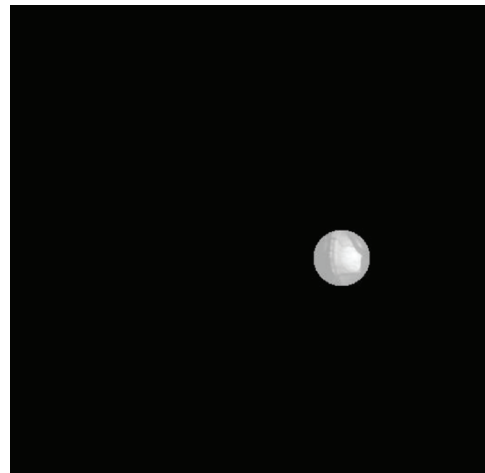


Fig. 5. Optic disc segmented image after blood vessels in-painted.

These equations are orthonormal basis of sub-spaces, which related to the resolution 2^j . Then, the wavelet atoms are determined by using scaling and three mother atoms: Ψ^h, Ψ^v and Ψ^d . These mother atoms calculated as the tensor product of one dimensional $\Psi(x)$ and $\varphi(x)$, which is represented in the Eqs. (6) and (7).

$$\varphi(x) = \varphi(x_1)\varphi(x_2), \quad \Psi^h(x) = \Psi(x_1)\Psi(x_2) \quad (6)$$

$$\Psi^v(x) = \varphi(x_1)\Psi(x_2), \quad \Psi^d(x) = \Psi(x_1)\varphi(x_2) \quad (7)$$

A two dimensional HDWT implemented using the combination of digital filter banks and down-samplers. The digital filter bank contains low-pass (h) and high-pass (g) filters. In the HDWT configuration structure, the number of banks set as per the desired resolution. As the image is a two dimensional, HDWT compute wavelet using separate wavelet functions. Then, the rows and columns of the glaucoma images are individually undergone through the one dimensional wavelet transform in order to establish the two dimensional HDWT.

The original glaucoma images $A_{2^{j+1}}f$ at resolution 2^{j+1} decomposed into four sub-band images in the frequency domain. Among four sub-band images, three sub-band images $D_{2^j}^h f$, $D_{2^j}^v f$, $D_{2^j}^d f$ are the glaucoma images at resolution 2^j in horizontal, vertical and diagonal directions. The fourth one is the approximation image, $A_{2^j}f$ found at coarse resolution, so, the whole glaucoma image $A_{2^{j+1}}f$ is represented in the Eq. (8).

$$A_{2^{j+1}}f = D_{2^j}^h f + D_{2^j}^v f + D_{2^j}^d f + A_{2^j}f \quad (8)$$

The decomposed glaucoma sub-images are the two dimensional orthogonal wavelet representation. The outcome of wavelet decomposition of a glaucoma image resultant into four orthogonal sub-bands: Low–Low (LL) band, Low–High (LH) band, High–Low (HL) band, and High–High (HH) band, which correspond to sub-images $D_{2^j}^h f$, $D_{2^j}^v f$, $D_{2^j}^d f$ and $A_{2^j}f$.

3.5. Hybrid Feature Extraction

In the proposed methodology, feature extraction performed on the wavelet transformed glaucoma images. Generally, feature extraction is defined as the action of mapping the image from image space to the feature space. In this scenario, a high level feature named as Gray-Level Co-occurrence Matrix (GLCM) employed for extracting the features of glaucoma image. GLCM is well known texture analysis technique that calculates the image characteristics related to second order statistics. Here, two effective GLCM texture features considered: homogeneity, and correlation. The brief description about GLCM features detailed below.

3.5.1. Homogeneity

Homogeneity measures the closeness of distribution elements in the gray level matrix. To quantitatively classify the homogeneous texture regions for similarity, the local spatial statistics of the texture calculated using scale and orientation selective of Gabor filtering. The glaucoma image subdivided into a set of homogeneous texture regions, then the texture features related to the regions of indexed image data. In GLCM, homogeneity calculates four direction (i.e., $\theta = 0^\circ, 45^\circ, 90^\circ$ or 135°) with a feature vector size of four. Homogeneity delivers high accuracy of detection in the defected areas, which described by a weak variation

in grey level. The general formula to calculate the homogeneity of glaucoma image represented in the Eq. (9).

$$\text{Homogeneity} = \sum_{i=0}^{n-1} \sum_{j=0}^{n-1} \frac{P_{ij}}{1 + (i - j)^2} \quad (9)$$

3.5.2. Correlation

Correlation technique calculates the linear dependency of grey levels of neighboring pixels. Glaucoma image correlation is an optical technique, which employs accurate two-dimension and three-dimension measurements of changes in glaucoma images. This is often used to measure deformation, displacement, strain and optical flow in the glaucoma image. Correlation widely applied in the field of medical science and engineering, which helps to detect the disorders in texture of glaucoma image. The general formula to estimate the correlation of glaucoma image is represented in the Eq. (9).

$$\text{Correlation} = \sum_{i=0}^{n-1} \sum_{j=0}^{n-1} P_{ij} \frac{(i - \mu)(j - \mu)}{\sigma^2} \quad (10)$$

Where, μ is represented as mean, σ is denoted as standard deviation, n is represented as number of gray levels of glaucoma image, $P(ij)$, represented as the pixel value of the position (i, j) of the glaucoma images, and P_{ij} is represented as normalized co-occurrence matrix.

3.6. Dimensionality Reduction Using Principal Component Analysis

After extracting the feature values using hybrid feature extraction, a statistical method: PCA used to decrease the dimension of extracted feature values. Various dimensionality reduction techniques preferred in glaucoma detection system, but the performance of PCA is very superior, because it factorize the target glaucoma region and sparse noises effectively. Compared to other dimensionality reduction methods, PCA has two key advantages: reduce the complexity while grouping the images, and have low noise sensitivity. Also, PCA gives more attention on covariance and variance structure of the new variables $x_1, x_2, x_3, x_4, \dots, x_p$. These variable magnitudes are much higher than the other variables, because these new variables receive heavy weights.

To avoid this reason, the variables are determined on scales with different ranges, otherwise the unit of the measurements are not equal. Let, R be the sample correlation matrix calculated from n observations for each principal component p of random variables. The Eigen-value and Eigen-vector pairs of RR is represented as $(\varepsilon_1, e_1), (\varepsilon_2, e_2), (\varepsilon_3, e_3), \dots, (\varepsilon_p, e_p)$. The feature vector size of fundus image is 100×6 (100 is the number of images and 6 is the coefficients of each image). The dimension of feature value size is reduced to 100×5 by applying PCA method. It is essential to convert a high dimensional image space into low dimensional image space in order to avoid the issue of “curse of dimensionality.” The i -th sample principal component of a vector $x = x_1, x_2, x_3, x_4, \dots, x_p$ variable is denoted in the Eq. (11).

$$e_i Z = e_{i1} Z_1 + e_{i2} Z_2 + e_{i3} Z_3 + \dots + e_{ip} Z_p, \quad i = 1, 2, 3, \dots, p \quad (11)$$

Where, $e_i Z = (e_{i1}, e_{i2}, \dots, e_{ip})$ represents i -th Eigen value and $Z = Z_1, Z_2, Z_3, \dots, Z_p$ $Z = Z_1, Z_2, Z_3, \dots, Z_p$ is the standardized vector observation.

Table I. Performance evaluation using HRF database.

		HRF database							
Dimensionality reduction	Classifiers	True positive (TP)	False positive (FP)	False negative (FN)	True negative (TN)	Negative predictive value (%)	Positive predictive value (%)	Sensitivity (%)	Specificity (%)
Without PCA	RF	1	1	1	1	50	50	50	50
	SVM	1	0	2	1	33	0	50	0
	NN	1	2	1	0	0	33	0	50
	NB	1	0	2	1	33	0	50	0
	KNN	2	1	0	1	100	66.7	50	100
With PCA	RF	1	0	2	1	33	0	50	0
	SVM	1	0	2	1	33	0	50	0
	NN	1	1	1	1	50	50	50	50
	NB	1	1	1	1	50	50	50	50
	KNN	2	0	0	2	100	100	100	100

In principal component, the sample variance is represented as $n_i n_j$ and the sample covariance pairs are mentioned as zero. In addition, the total sample variance in all standardized variables is equal to the total sample variance in the principal component. Mathematically, the standardized vector observation is stated in the Eq. (12).

$$Z_k = \frac{x_k - \bar{x}_k}{\sqrt{v_{kk}}}, \quad k = 1, 2, 3, \dots, p \quad (12)$$

Where, x_k is denoted as sample mean and v_{kk} is denoted as sample variance of the variable x_k .

3.7. Classification

After performing dimensionality reduction, the classification is carried-out on the optimal feature values. Classification is defined as a boundary between the classes in order to label the classes based on their measured features. In this scenario, KNN classifier is undertaken for classifying the normal and abnormal glaucoma cases. If there is no prior knowledge about the distribution of the data, the KNN method should be one of the first choices for classification.

3.7.1. K-Nearest Neighbor

In recent years, KNN is one of the emerging classification methods with the advantage of limited usage of resources, while training the large dataset. KNN is an un-supervised learning classification algorithm and also one of the most widely used non-parametric pattern classification method, which reduce the concern of complexity of probability density. The KNN rule classifies the x by allocating the label frequently in K -nearest samples. Initially, KNN methodology trains the record and then identifies the new records, which are similar to the trained records. It searches the space for k training records that are nearest to the new record as the new record neighbors. In this algorithm, the nearest distance is defined by means of a distance metric named as Euclidean distance. Euclidean distance between the two records (or two points in n -dimensional space) is defined in the Eq. (13)

$$X_1 = (X_{11}, X_{12}, \dots, X_{1n}), \quad X_2 = (X_{21}, X_{22}, \dots, X_{2n}), \quad (13)$$

$$\text{dist}(x_1, x_2) = \sqrt{\sum_{i=1}^n (x_{1i} - x_{2i})^2}$$

Where, x_1 and x_2 are two records with n attributes. This formula measures the distance between x_1 and x_2 based on the difference between the corresponding values of the attribute in record x_1 and x_2 .

4. RESULTS AND DISCUSSION

For experimental simulation, MATLAB (version 2017a) was employed on PC with 3.2 GHz with i5 processor. The performance of proposed methodology was compared with NN, SVM, NB and RF in terms of specificity, accuracy, sensitivity, PPV, NPV and MCC on the reputed glaucoma datasets: HRF and RIMONE to estimate the efficiency of the proposed methodology.

4.1. Performance Evaluation

The relationship between the input and output variables of a system understand by employing the suitable performance metrics like sensitivity and specificity. The general formula for calculating the sensitivity and specificity of normality and abnormality of glaucoma detection rate is given in the Eqs. (14) and (15).

$$\text{Sensitivity} = \frac{\text{Number of TP}}{\text{Number of TP} + \text{Number of FN}} \times 100 \quad (14)$$

$$\text{Specificity} = \frac{\text{Number of TN}}{\text{Number of TN} + \text{Number of FP}} \times 100 \quad (15)$$

In addition, accuracy, PPV, NPV and MCC are the suitable evaluation metrics for finding the effectiveness of normality and abnormality of glaucoma detection rate. These measures are statistical variability and a description of random errors. The general formula of accuracy, PPV, NPV and MCC for determining the normality and abnormality of glaucoma detection rate is represented in the Eqs. (16)–(19).

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \times 100 \quad (16)$$

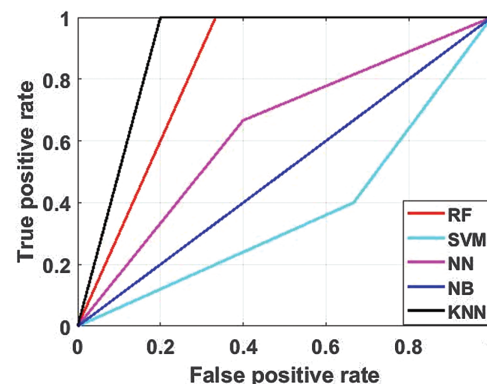


Fig. 6. Graphical comparison of HRF database.

Table II. Performance evaluation using RIMONE database.

		RIMONE database							
Dimensionality reduction	Classifiers	True positive (TP)	False positive (FP)	False negative (FN)	True negative (TN)	Negative predictive value (%)	Positive predictive value (%)	Sensitivity (%)	Specificity (%)
Without PCA	RF	10	5	0	5	100	66.7	50	100
	SVM	10	6	0	4	100	62	40	100
	NN	5	6	5	4	44	45.45	50	40
	NB	6	6	4	4	50	50	40	60
	KNN	10	2	0	8	100	83	80	100
With PCA	RF	10	1	0	9	100	90.90	90	100
	SVM	10	3	0	7	100	76.92	70	100
	NN	8	5	2	5	71.42	61.53	50	80
	NB	9	0	1	9	90.90	100	100	90
	KNN	10	0	0	10	100	100	100	100

$$PPV = \frac{TP}{TP + FP} \times 100 \quad (17)$$

$$NPV = \frac{TN}{TN + FP} \times 100 \quad (18)$$

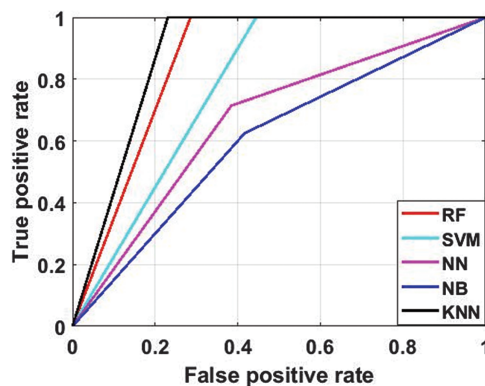
$$MCC = \frac{TP \times TN - FP \times FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \times 100 \quad (19)$$

Where, TP represents true positive, FP denotes false positive, TN presents true negative and FN specified as false negative.

4.2. Result for HRF Database

In this experimental analysis, HRF database assessed for comparing the performance evaluation of existing methods and the proposed method. For experimental validation, random testing and training of image occurred with 50 number of iterations. In Table I, the sensitivity of the proposed technique: KNN delivered 100% and the existing methodologies: RF, SVM, NN and NB delivered 50%, 50%, 50% and 50% of sensitivity. Similarly, the specificity of the proposed technique delivered 100% and the existing methodologies delivered 0%, 0%, 50% and 50% of specificity.

Likewise, the PPV of the proposed technique delivered 100% and the existing methodologies delivered 0%, 0%, 50% and 50% of PPV. Similarly, the NPV of proposed methodology achieved 100% and the existing methodologies delivered 33%, 33%, 50% and 50% of NPV. The graphical comparison of HRF database presented in the Figure 6.

**Fig. 7.** Graphical comparison of RIMONE database.

4.3. Result for RIMONE Database

In this section, RIMONE database considered in comparing the performance evaluation of existing methods and the proposed scheme. In Table II, the sensitivity of the proposed technique: KNN delivered 100% and the existing methodologies: RF, SVM, NN and NB delivered 90%, 70%, 50% and 100% of sensitivity.

Table III. Accuracy comparison of proposed and existing methodologies.

Database	Classification method	Feature extraction method	Feature reduction method	Accuracy (%)
HRF	RF	Homogeneity and correlation	PCA	80
	SVM	Homogeneity and correlation	PCA	70.2
	NN	Homogeneity and correlation	PCA	52.5
	NB	Homogeneity and correlation	PCA	61.2
	KNN	Homogeneity and correlation	PCA	85
RIMONE	RF	Homogeneity and correlation	PCA	93.4
	SVM	Homogeneity and correlation	PCA	73.4
	NN	Homogeneity and correlation	PCA	77.5
	NB	Homogeneity and correlation	PCA	86.3
	KNN	Homogeneity and correlation	PCA	99

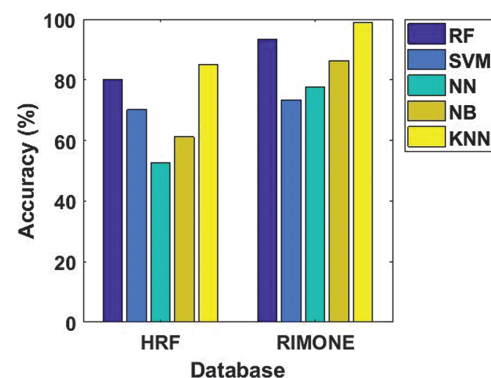
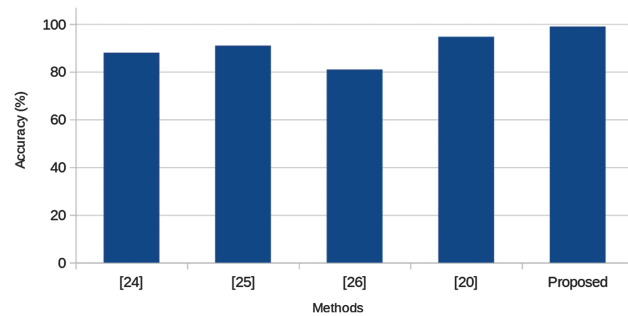
**Fig. 8.** Graphical representation of accuracy comparison.

Table IV. Cross validation of proposed methodology.

Database	HRF	RIMONE
Feature extraction method	Homogeneity and correlation	Homogeneity and correlation
Feature reduction method	PCA	PCA
Classification method	KNN	KNN
Total number of samples	20	100
Training samples	16	80
Testing samples	4	20
Number of glaucoma cases	10	50
Number of normal cases	10	50
True positive	50	50
False negative	1	1
Accuracy (%)	85	99
Specificity (%)	81	100
Sensitivity (%)	88	98.04
Positive predictive value (%)	80	100
Negative predictive value (%)	90	98
MCC (%)	98.01	70.35

Similarly, the specificity of the proposed technique delivered 100% and the existing methodologies delivered 100%, 100%, 80% and 90% of specificity. Correspondingly, the PPV of proposed technique delivered 100% and the existing methodologies delivered 90.90%, 76.92%, 61.53% and 100% of PPV. Similarly, the NPV of proposed methodology achieved 100% and the existing methodologies delivered 100%, 100%, 71.42% and 90.90% of NPV. Tables I and II shows the performance evaluation of existing methodologies and the proposed method. Here, the proposed approach captures the non-linear characteristics of the fundus images and preserve quantitative relationships between the low level and high level feature variables. The evaluation metrics confirms that the proposed scheme performs significantly in Glaucoma detection compared to previous methods in terms of specificity, sensitivity, accuracy, PPV, NPV and MCC. The graphical comparison of RIMONE presented in the Figure 7.

Table III presented the performance of various classifiers on hybrid feature extraction for improving the accuracy of classification. Prominent features selected using hybrid feature extraction

**Fig. 9.** Graphical representation of comparative analysis.

technique and PCA, only the selected features considered for classification. Table III clearly showed that the KNN approach improved the accuracy in glaucoma detection up to 3–30% compared to the existing classification methods in the both HRF and RIMONE. The graphical representation of accuracy comparison represented in the Figure 8.

Table IV showed the outcome of cross validation using KNN. This cross validation approach applied for two databases: HRF and RIMONE. The KNN approach delivered an accuracy of 85%, 99% with specificity of 81%, 100% and sensitivity of 88%, 98.04%. The positive predictive and negative predictive value for glaucoma classification is 80%, 100% and 90%, 98% respectively. The MCC is also calculated for both the databases, HRF database achieved 98.01% and RIMONE database delivered 70.35%. The result of KNN showed that the proposed glaucoma detection method performed well and also provided a significant contribution in computer-aided health monitoring system.

4.4. Comparative Analysis

Table V presented the comparative study of existing work and the proposed work performance. Kim et al.²¹ developed an automatic glaucoma detection system using multi-fractional Brownian motion approach and box-counting methodology. After obtaining the feature values, Gaussian kernel based multi-class classification approach was employed for classification. The developed feature extraction based approach obtained 88% of classification accuracy. Additionally, Acharya et al.²² presented an automatic method using the combination of high order spectra

Table V. Comparative analysis of proposed and existing methodologies.

References	Database	Feature considered	Feature selection or reduction	Classification method	Accuracy (%)
[24]	I-D retinal nerve fiber layer	Seven features including fractal analysis (FA) features.	PCA	SVM	88
[25]	Fundus image	Twenty features are utilized with the combination of texture and higher order spectra features.	Gain ratio, chi squared, and Information gain	RF	91
[26]	Fundus image	Three feature pixel intensity values, FFT and B-spline coefficients.	PCA	SVM	81
[20]	Segmented optic disc fundus image	Eighteen features considered from first level wavelet features.	PCA	SVM and KNN	94.7
Proposed	RIMONE	Hybrid features (correlation and homogeneity)	PCA	KNN	99

and texture features. The obtained feature vectors were classified by employing RF classifier. This experiment was carried out on fundus image to validate its result in terms of accuracy and achieved 91%.

Nyúl,²³ developed PCA feature extraction for compressing the high dimensional vectors and then SVM approach was used for classification. The developed technique achieved 81% of classification accuracy. In addition, Singh et al.²⁰ proposed wavelet feature extraction followed by optimized genetic feature selection. This work was based on feature extraction from the segmented blood vessels and optic disc for improving the detection accuracy. The experimental outcome showed that the developed methodology achieved 94.7% of classification accuracy. Whereas, the proposed work achieved 99% of accuracy that was higher than the existing works. Additionally, validate the result for another one dataset named as HRF database and attained 85% of accuracy in classification. The graphical representation of comparative analysis presented in the Figure 9.

5. CONCLUSION

Glaucoma disease detection is one of the most significant research tasks in computer-aided health monitoring system. The objective of the experiment is to develop a proper segmentation and feature extraction approach for classifying the normality and abnormality of glaucoma disease using HRF and RIMONE databases. In this scenario, correlation based template matching segmentation used to segment the optic discs. Then, hybrid feature extraction (homogeneity and correlation) employed for attaining optimal feature subsets along with PCA scheme in order to eliminate the irrelevant feature vectors. The obtained feature values classified using the classifier: KNN classification method. Associated with other obtainable approaches in glaucoma detection, the proposed scheme delivered an effective performance by means of accuracy, sensitivity, specificity, PPV, NPV and MCC, around 3–30% enhancement than the existing methods. In the future work, for further improving the glaucoma detection rate, the descriptor level features are combined with an appropriate multi-objective classifier.

References and Notes

1. M. Soorya, A. Issac, and M. K. Dutta, An automated and robust image processing algorithm for glaucoma diagnosis from fundus images using novel blood vessel tracking and bend point detection. *International Journal of Medical Informatics* 110, 52 (2018).
2. S. Roychowdhury, D. D. Koozekanani, and K. K. Parhi, Blood vessel segmentation of fundus images by major vessel extraction and sub image classification. *IEEE Journal of Biomedical and Health Informatics* 19, 1118 (2015).
3. J. A. De La Fuente-Arriaga, E. M. Felipe-Riverón, and E. Garduño-Calderón, Application of vascular bundle displacement in the optic disc for glaucoma detection using fundus images. *Computers in Biology and Medicine* 47, 27 (2014).
4. S. Maheshwari, R. B. Pachori, and U. R. Acharya, Automated diagnosis of glaucoma using empirical wavelet transform and correntropy features extracted from fundus images. *IEEE Journal of Biomedical and Health Informatics* 21, 803 (2017).
5. L. C. Neto, G. L. Ramalho, J. F. R. Neto, R. M. Veras, and F. N. Medeiros, An unsupervised coarse-to-fine algorithm for blood vessel segmentation in fundus images. *Expert Systems with Applications* 78, 182 (2017).
6. G. D. Joshi, J. Sivaswamy, and S. R. Krishnadas, Optic disk and cup segmentation from monocular color retinal images for glaucoma assessment. *IEEE Transactions on Medical Imaging* 30, 1192 (2011).
7. A. Belghith, M. Balasubramanian, C. Bowd, R. N. Weinreb, and L. M. Zangwill, A unified framework for glaucoma progression detection using heidelberg retina tomograph images. *Computerized Medical Imaging and Graphics* 38, 411 (2014).
8. S. Patyal, A. Kotwal, A. Banarji, and V. S. Gurunadh, Frequency doubling technology and standard automated perimetry in detection of glaucoma among glaucoma suspects. *Medical Journal Armed Forces India* 70, 332 (2014).
9. J. M. Miguel-Jiménez, L. Boquete, S. Ortega, J. M. Rodríguez-Ascariz, and R. Blanco, Glaucoma detection by wavelet-based analysis of the global flash multifocal electroretinogram. *Medical Engineering and Physics* 32, 617 (2010).
10. P. V. Rao, R. Gayathri, and R. Sunitha, A novel approach for design and analysis of diabetic retinopathy glaucoma detection using cup to disk ration and ANN. *Procedia Materials Science* 10, 446 (2015).
11. R. Bock, J. Meier, L. G. Nyúl, J. Horneberger, and G. Michelson, Glaucoma risk index: Automated glaucoma detection from color fundus images. *Medical Image Analysis* 14, 471 (2010).
12. Y. H. Hwang, Y. C. Jeong, H. K. Kim, and Y. H. Sohn, Macular ganglion cell analysis for early detection of glaucoma. *Ophthalmology* 121, 1508 (2014).
13. D. S. Chang, K. S. Arora, M. V. Boland, W. Supakontanasan, and D. S. Friedman, Development and validation of an associative model for the detection of glaucoma using pupillography. *American Journal of Ophthalmology* 156, 1285 (2013).
14. C. Bowd, G. Vizzeri, A. Tafreshi, L. M. Zangwill, P. A. Sample, and R. N. Weinreb, Diagnostic accuracy of pattern electroretinogram optimized for glaucoma detection. *Ophthalmology* 116, 437 (2009).
15. C. Bowd, L. M. Zangwill, R. N. Weinreb, F. A. Medeiros, and A. Belghith, Estimating optical coherence tomography structural measurement floors to improve detection of progression in advanced glaucoma. *American Journal of Ophthalmology* 175, 37 (2017).
16. P. S. Mittapalli and G. B. Kande, Segmentation of optic disk and optic cup from digital fundus images for the assessment of glaucoma. *Biomedical Signal Processing and Control* 24, 34 (2016).
17. A. A. Salam, T. Khalil, M. U. Akram, A. Jameel, and I. Basit, Automated detection of glaucoma using structural and non-structural features. *Springer-Plus* 5, 1519 (2016).
18. S. Roychowdhury, D. D. Koozekanani, S. N. Kuchinka, and K. K. Parhi, Optic disc boundary and vessel origin segmentation of fundus images. *IEEE Journal of Biomedical and Health Informatics* 20, 1562 (2016).
19. S. Maheshwari, R. B. Pachori, V. Kanhangad, S. V. Bhandary, and U. R. Acharya, Iterative variational mode decomposition based automated detection of glaucoma using fundus images. *Computers in Biology and Medicine* 88, 142 (2017).
20. A. Singh, M. K. Dutta, M. P. Sarathi, V. Uher, and R. Burget, Image processing based automatic diagnosis of glaucoma using wavelet features of segmented optic disc from fundus image. *Computer Methods and Programs in Biomedicine* 124, 108 (2016).
21. S. H. Wang, P. Phillips, Z. C. Dong, and Y. D. Zhang, Intelligent facial emotion recognition based on stationary wavelet entropy and jaya algorithm. *Neurocomputing* 272, 668 (2018).
22. Y. Zhang, Z. Dong, A. Liu, S. Wang, G. Ji, Z. Zhang, and J. Yang, Magnetic resonance brain image classification via stationary wavelet transform and generalized eigenvalue proximal support vector machine. *Journal of Medical Imaging and Health Informatics* 5, 1395 (2015).
23. Z. Lu, S. Lu, G. Liu, Y. Zhang, J. Yang, and P. Phillips, A pathological brain detection system based on radial basis function neural network. *Journal of Medical Imaging and Health Informatics* 6, 1218 (2016).
24. P. Y. Kim, K. M. Iftekharuddin, P. G. Davey, M. Tóth, A. Garas, G. Holló, and E. A. Essock, Novel fractal feature-based multiclass glaucoma detection and progression prediction. *IEEE Journal of Biomedical and Health Informatics* 17, 269 (2013).
25. U. R. Acharya, S. Dua, X. Du, and C. K. Chua, Automated diagnosis of glaucoma using texture and higher order spectra features. *IEEE Transactions on Information Technology in Biomedicine* 15, 449 (2011).
26. L. G. Nyúl, Retinal image analysis for automated glaucoma risk evaluation, *Proceedings of the MIPPR: Medical Imaging, Parallel Processing of Images, and Optimization Techniques* (2009) p. 7497.

Received: 2 April 2018. Accepted: 4 June 2018.